



جامعة الكفيل
University of Alkafeel

Oral & Maxillofacial Radiology

Extra Oral Radiography

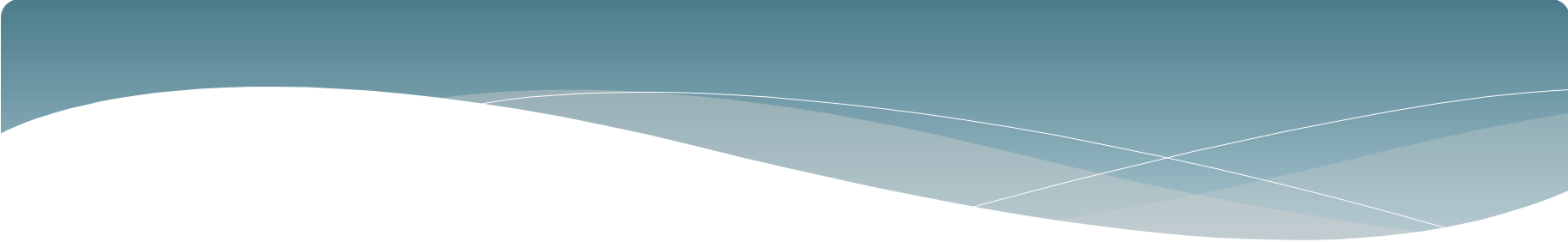
ا.م. سرمد محمد حموزي

A. prof. Sarmad Hamozi

B.D.S, M.Sc.,OMFR, Ortho ADA

dr.sarmadh@alkafeel.com

Extraoral radiographs are used to examine areas not fully covered by intraoral radiographs or to evaluate (the cranium, face including the maxilla and mandible, or cervical spine) for diseases, trauma, abnormalities, relationship between various orofacial and dental structures, growth and development of the face, or treatment progression.



1. Craniofacial projections

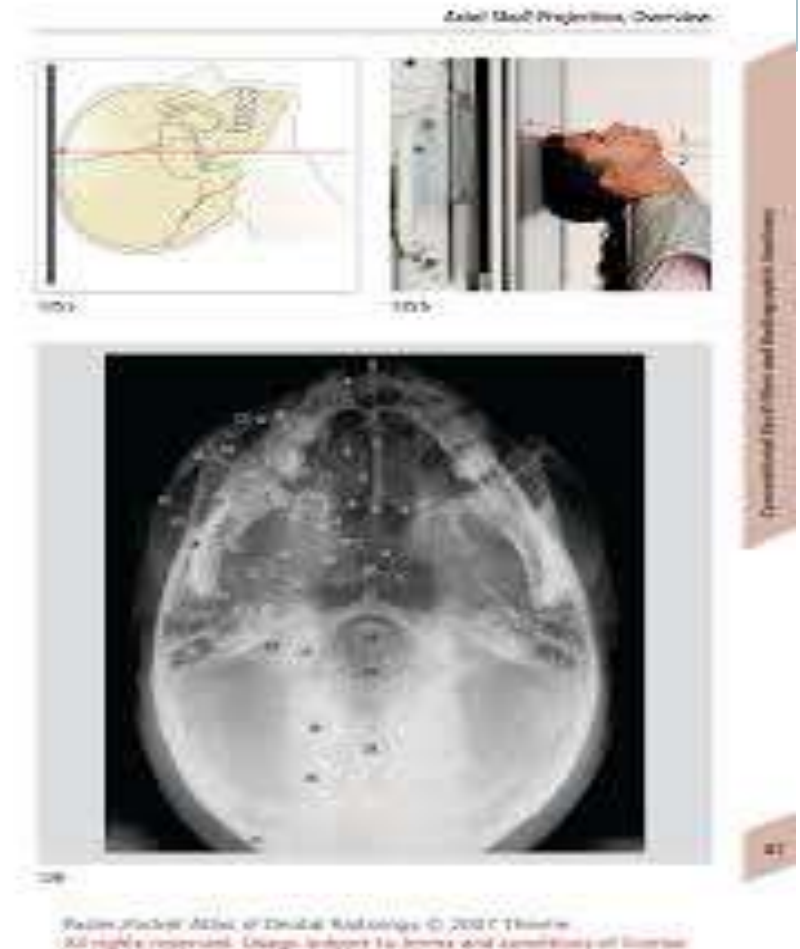
2. Cephalometric projections

3. TMJ projections

1. Craniofacial and skull projections

a. Submentovertex (SMV) or Base Projection

- * In this projection the patient's neck is extended as far backward as possible and the central beam is perpendicular to the image receptor, directed from below the mandible toward the vertex of the skull (so named *SMV*),

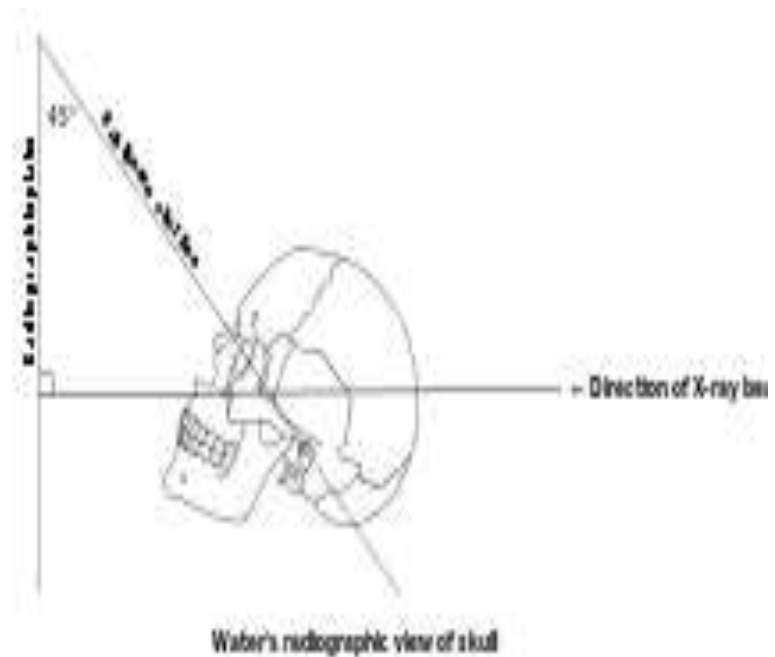


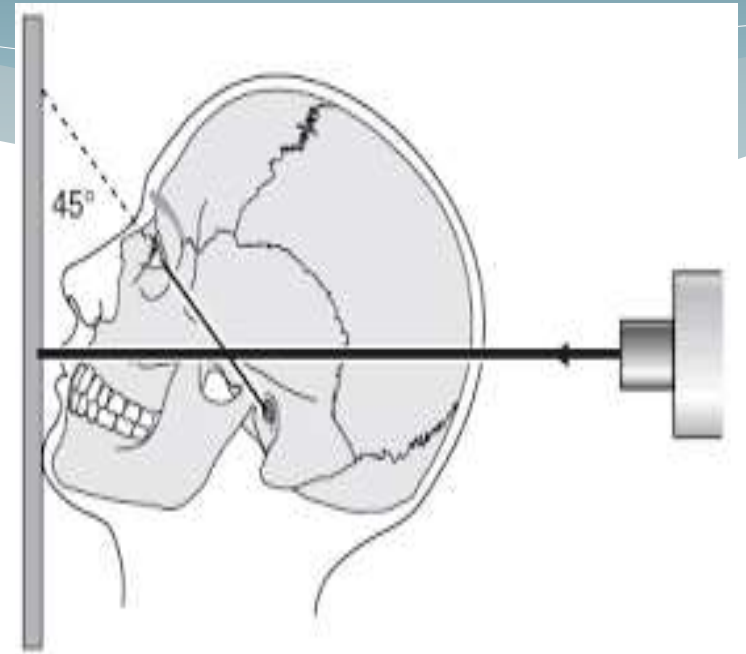


it displays the base of the skull, the zygomatic arches fracture, and the sphenoid sinuses, hard palate, and mandible.

b.Waters (Occipitomenatal) Projection

- * It is the best projection to view 'sinuses'. The image receptor is placed in front of the patient and the central beam is perpendicular to the image receptor and centered in the area of the maxillary sinuses.





It displays predominantly the **maxillary sinuses**. It also images the **orbital ridges and floor, frontal and ethmoid sinuses** as well as the **nasal cavity and midfacial fractures**.

c. Reverse Towne Projection

- * It is an open mouth technique, the image receptor is placed in front of the patient, The central beam is perpendicular to the image receptor and parallel to the patient's midsagittal plane centered at the level of the condyles,



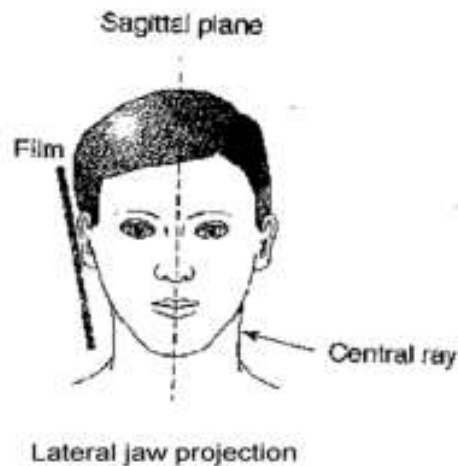
Reverse Towne Projection

used to evaluate patients with suspected fractures of **the condyle and condylar neck** also show the **posterolateral wall of the maxillary antrum**.



d. Lateral Oblique Projection

An extraoral view of the jaw used to assess presence of unerupted teeth, fractures of the mandible, and evaluation of jaw lesions,



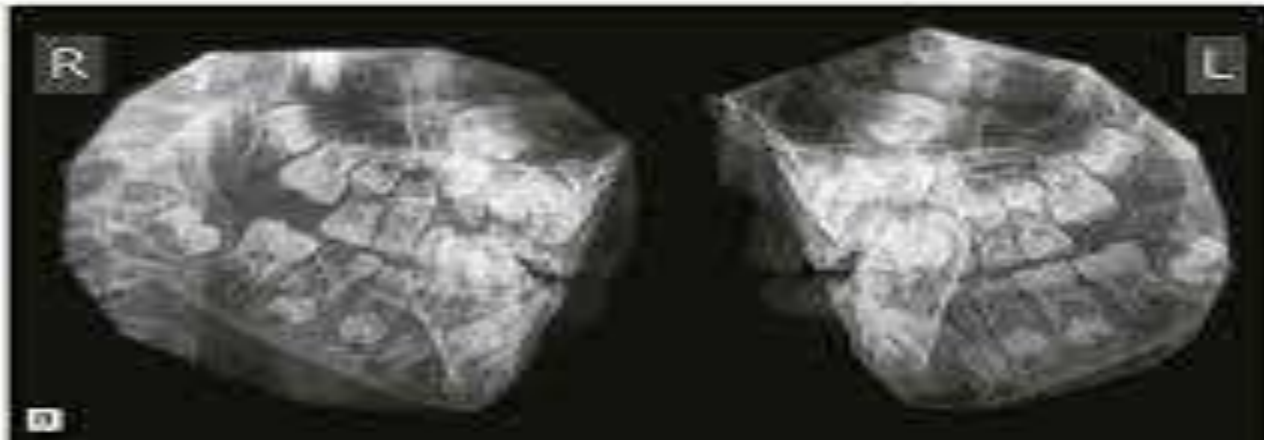
Lateral Oblique Projection

- * **1. Mandibular ramus projection** it gives a view of the ramus from the angle of the mandible to the condyl. It often is very useful for examining the third molar regions of the maxilla and
- * **2. Mandibular body projection** it demonstrates the premolar-molar region and the inferior border of the mandible. It provides much broader coverage than is possible with periapical projections



LATERAL OBLIQUE RADIOGRAPH

- 3. Bimolar view: In this view, both the right and left sides of mandible are taken in one film.

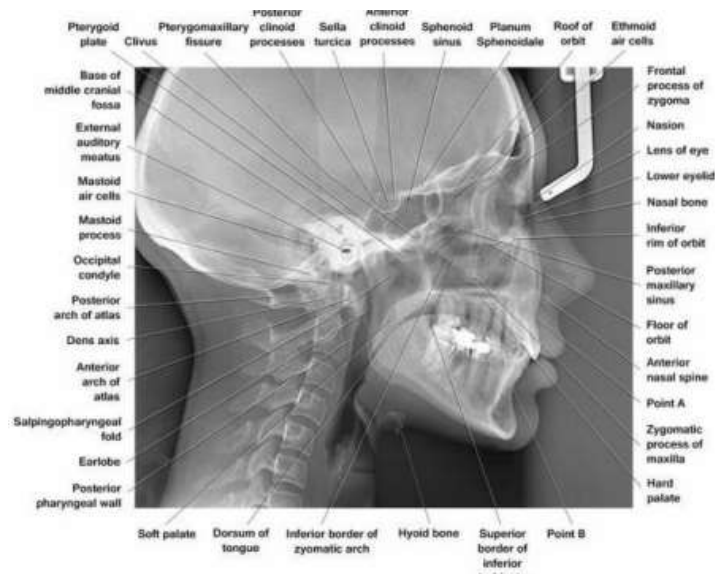


Cephalometric projections

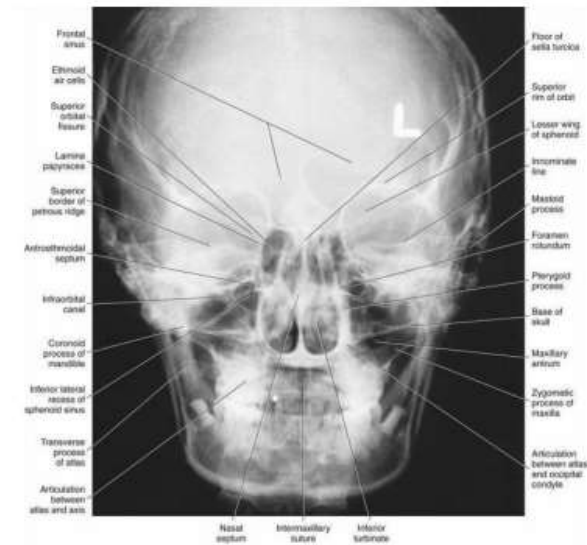
- * They are standardized projections made with a Cephalostat, which helps to maintain a **constant relationship between the skull, the receptor, and the x-ray beam.**
- * A cephalometric projection used a long **source-to-object distance of 5 ft; to minimizes image magnification.** The **object-to-receptor distance is typically 10 to 15 cm.**
- * These projections using film or digital receptors.

Cephalometric projections

1. Lateral Cephalometric Projection (Lateral Skull Projection)



2. Postero-anterior PA cephalometric projection



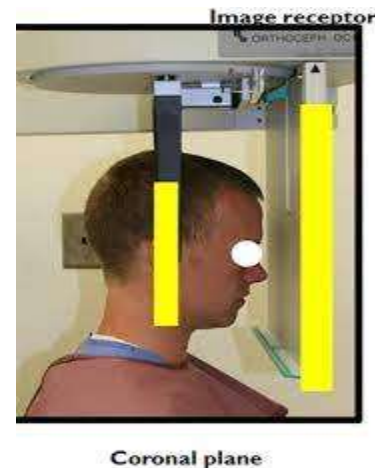
Lateral Cephalometric Projection (Lateral Skull Projection)

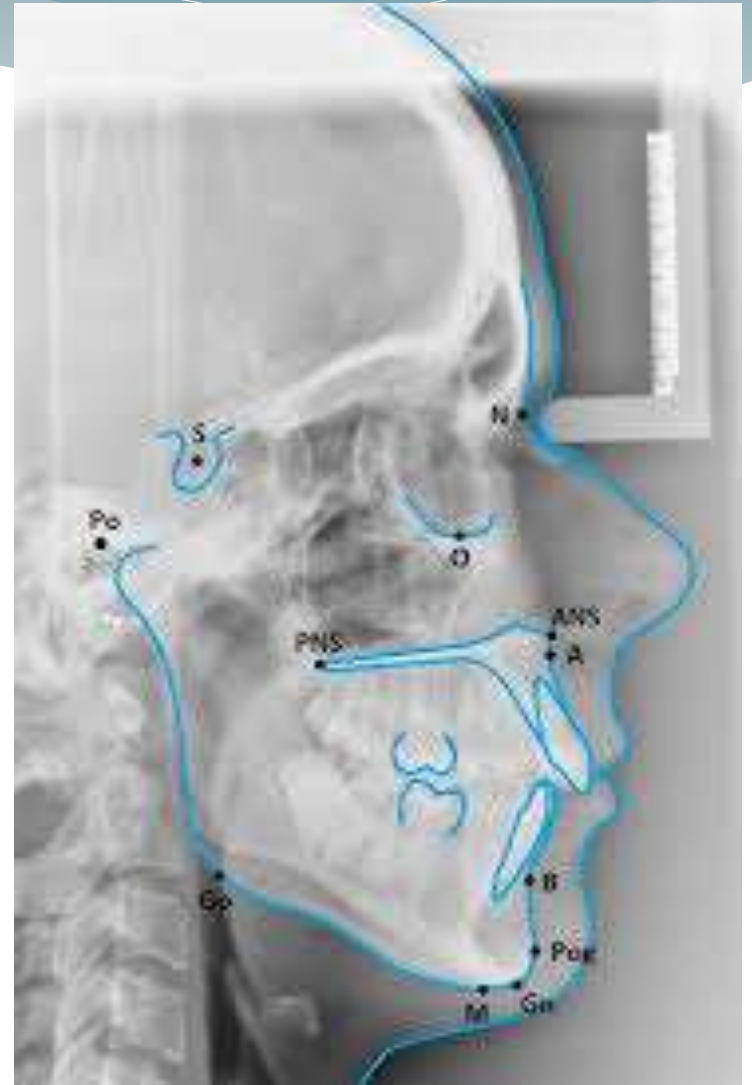


- * **Technique and positioning:**
- * The image receptor is positioned parallel to the patient's midsagittal plane
- * patient is placed with the left side toward the image receptor and asked to occlude in his normal intercuspation position.
- * The central beam is perpendicular to the midsagittal plane and the image receptor centered over the external auditory meatus.
- * **Exact superimposition of right and left sides is impossible because the structures on the side near to the image receptor are magnified less than the same structures on the side far from the image receptor.**

Postero-anterior cephalometric projection

- * **Technique and positioning:**
- * The image receptor is placed in front of the patient, perpendicular to the midsagittal plane and parallel to the coronal plane.
- * The central beam is perpendicular to the image receptor, directed from posterior to anterior (hence the name PA) at the level of the bridge of the nose.





Most common Cephalometric Landmarks

Skeletal Landmarks

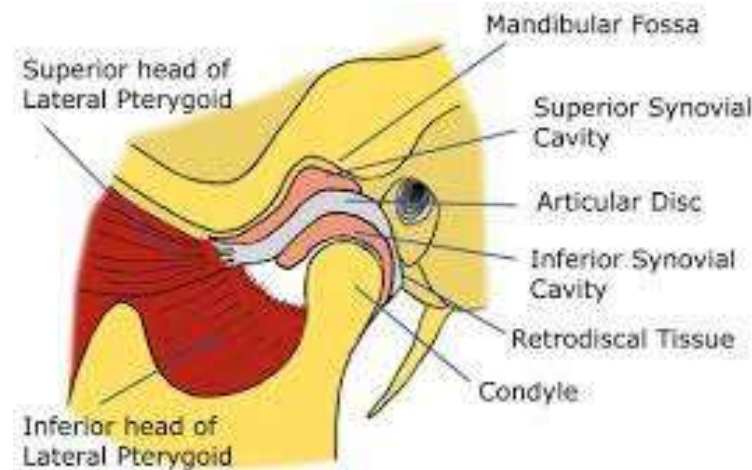
1. Porion (P): Most superior point of the external auditory canal
2. Sella (S): Center of the hypophyseal fossa
3. Nasion (N): Frontonasal suture
4. Orbitale (O): Most inferior point of the infraorbital rim
5. PNS: Tip of the posterior nasal spine
6. ANS: Tip of the anterior nasal spine
7. A point (A): Deepest point of the anterior border of the maxillary alveolar ridge concavity
8. B point (B): Deepest point in the concavity of anterior border of the mandible
9. Pogonion (Po): Most anterior point of the symphysis
10. Gnathion: Midpoint of the symphysis outline between pogonion and menton
11. Menton (M): Most inferior point of the symphysis
12. Gonion: Most convex point along the inferior border of the mandibular ramus

Soft Tissue Landmarks

1. Soft tissue glabella: Most anterior point of the soft tissue covering the frontal bone
2. Soft tissue nasion: Most concave point of soft tissue outline at the bridge of the nose
3. Tip of nose: Most anterior point of the nose
4. Subnasale: Soft tissue point where the curvature of the upper lip connects to the floor of the nose
5. Soft tissue A point: Most concave point of the upper lip between the subnasale and the upper lip point
6. Upper lip: Most anterior point of the upper lip
7. Lower lip: Most anterior point of the lower lip
8. Soft tissue B point: Most concave point of the lower lip between the chin and lower lip point
9. Soft tissue pogonion: Most anterior point of the soft tissue of the chin
10. Soft tissue gnathion: Midpoint of the chin soft tissue outline between the soft tissue pogonion and soft tissue menton
11. Soft tissue menton: Most inferior point of the soft tissue of the chin

Tempromandibular Joint (TMJ) projections

- * TMJ is the area where the mandible articulates with cranium. TMJ is consists of : Bones (mandibular condyle and temporal bone), Articular disc, Capsular ligament, and Joint cavity (synovial membrane)



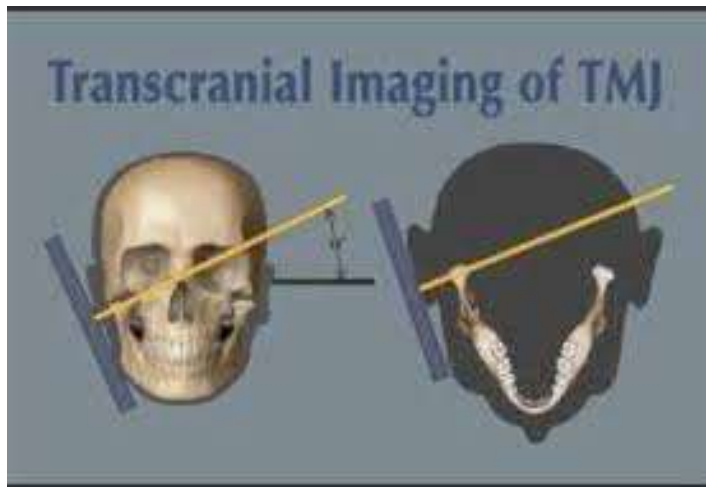
The Temporomandibular Joint

Types of imaging techniques

- * 1. Plain radiographs (transcranial projection, transorbital projection, transpharyngeal projection)
- * 2. Panoramic radiography (OPG)
- * 3. Cone beam computed tomography (CBCT) and Multidetector computed tomography (MDCT)
- * 4. MRI and ultrasonography (for soft tissue imaging)

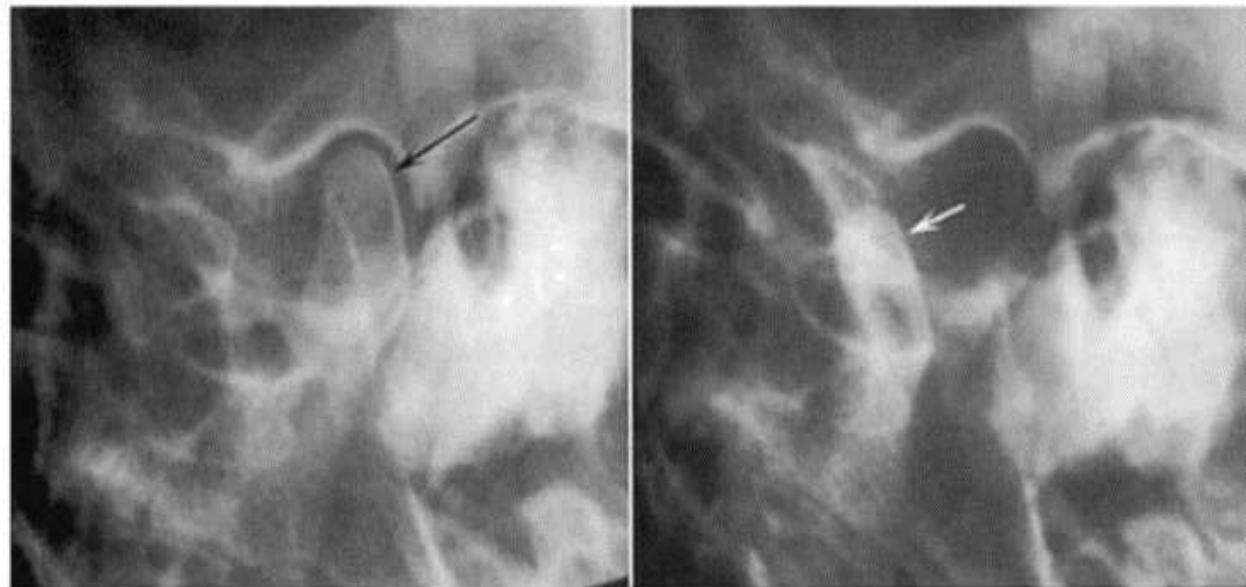
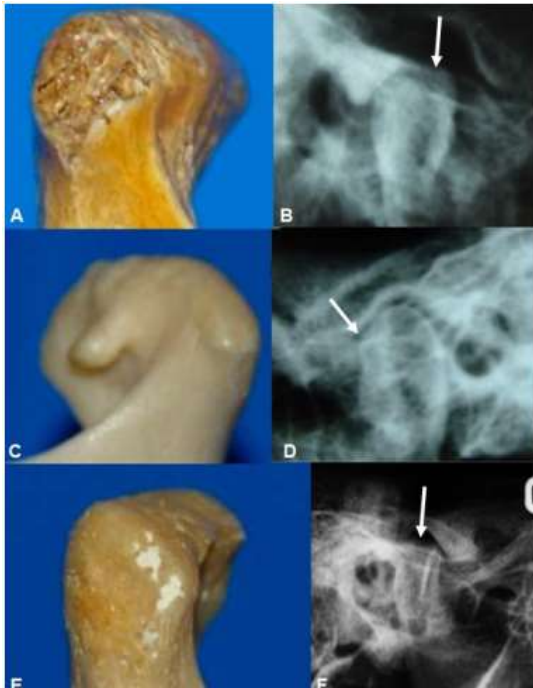
Transcranial-view

it is one of the **most popular plain imaging techniques to study the TMJ**. The film cassette is positioned against the facial skin surface on the side of interest and the central beam projected across the cranium passing through the TMJ from other side **either with Closed-mouth or Openmouth position**



Transcranial view

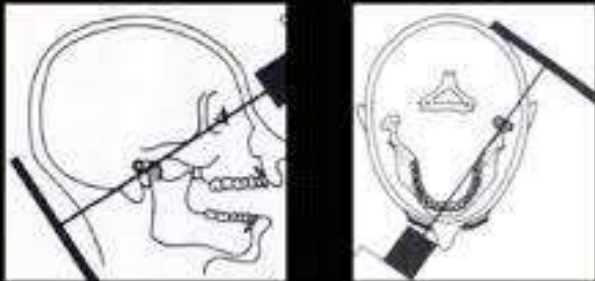
It used in visualization of changes in lateral aspect of the articulating surface, position and shape of condyle and fossa.



Transorbital view:

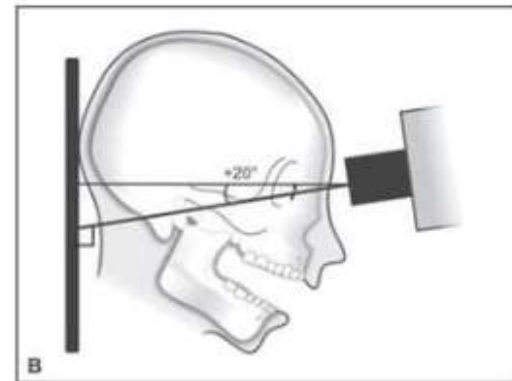
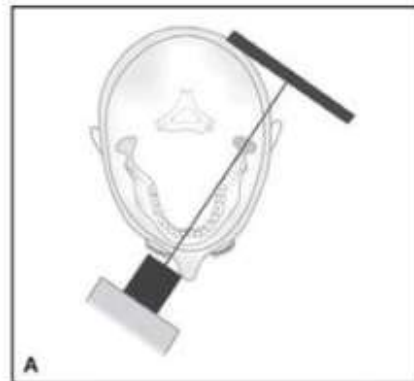
is a frontal projection of the TMJ. the film cassette Position behind the patient's head so that the central ray is projected infront of the patient through the orbit to the TMJ asking the patient to open the mouth.

Transorbital view



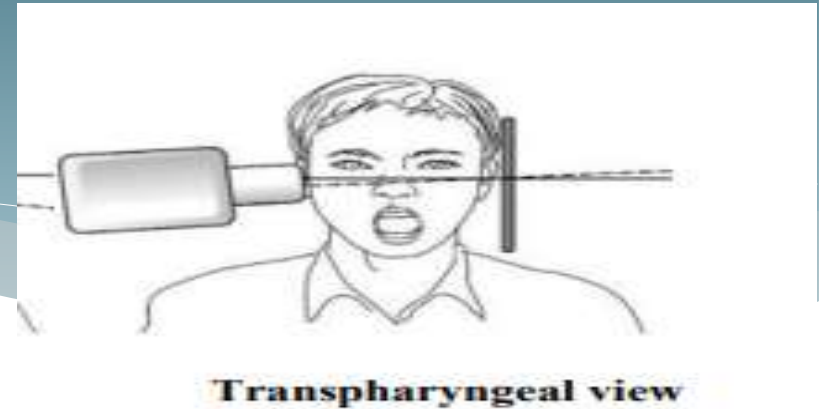
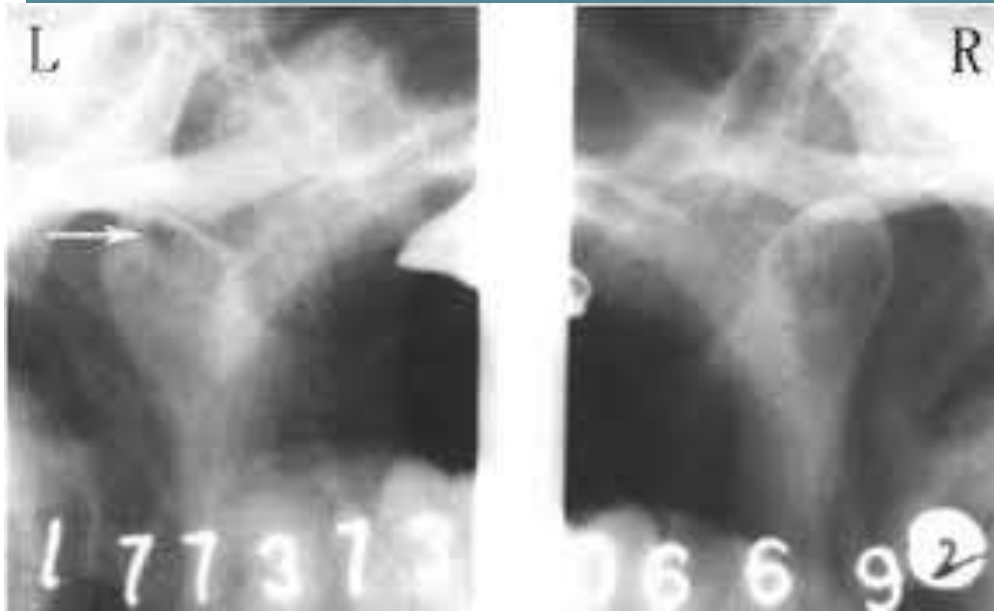
Transorbital view

It demonstrate the entire latero-medial articulating surface of the condyle and articulator eminence and the condylar neck.

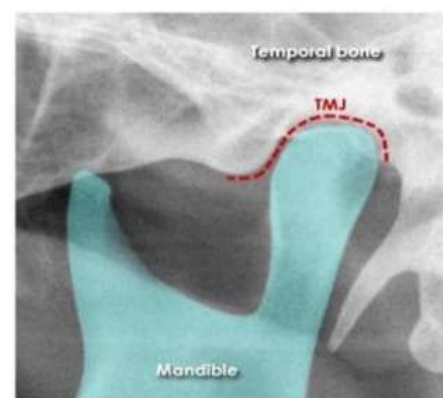


Transpharyngeal view

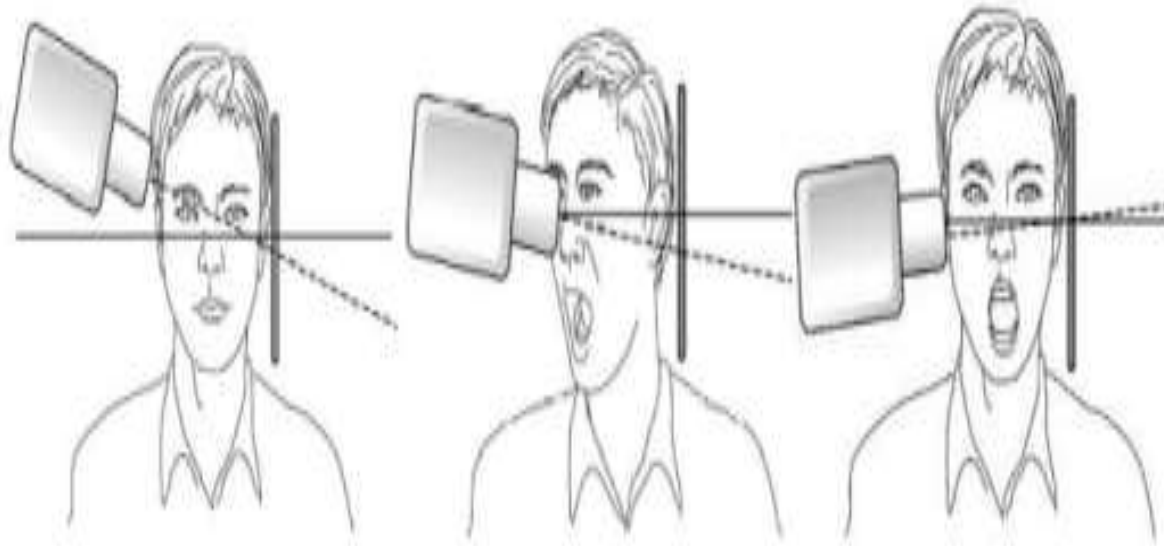
- * It also called infracranial view, the film cassette is positioned against the side of patient's head, parallel to the sagittal plane next to TMJ of interest. the central X-ray directs below the base of the skull through the oropharynx to TMJ, **before film exposure the patient opens the mouth wide.**



- * it demonstrate the angular process, diagnosis of fracture in the condyle and condylar neck, detection the alteration in condyle morphology.



Plain radiographs



Transcranial view

Transorbital view

Transpharyngeal view

Abnormalities of the TMJ

- * 1- Developmental abnormalities such as (condylar hypoplasia, condylar hyperplasia and coronoid hyperplasia, bifid condyle)
- * 2- Soft tissue abnormalities such as disc displacement (with or without reduction)
- * 3- Remodeling and arthritic conditions
- * 4- Trauma such as fracture, dislocation and effusion
- * 5- Ankylosis such as bony and soft tissue ankylosis
- * 6- Tumors (benign and malignant)

